

Tusk Robots E-Series (Invisible Fork Arm) Pallet AMR - CAD Design & Dimensioning Guide

This document provides all verified dimensions, mechanical specifications, tolerances, and CAD blueprint sheets required to 3D model and manufacture the **Tusk Robots E-Series Low-Profile Pallet AMR** (E10 Standard & E10T Telescopic "Invisible Fork Arm" models).

1. Master CAD Dimensioning Summary

Below is the consolidated master dimensions table to be used directly in SolidWorks, Fusion 360, or Inventor for 3D modeling:

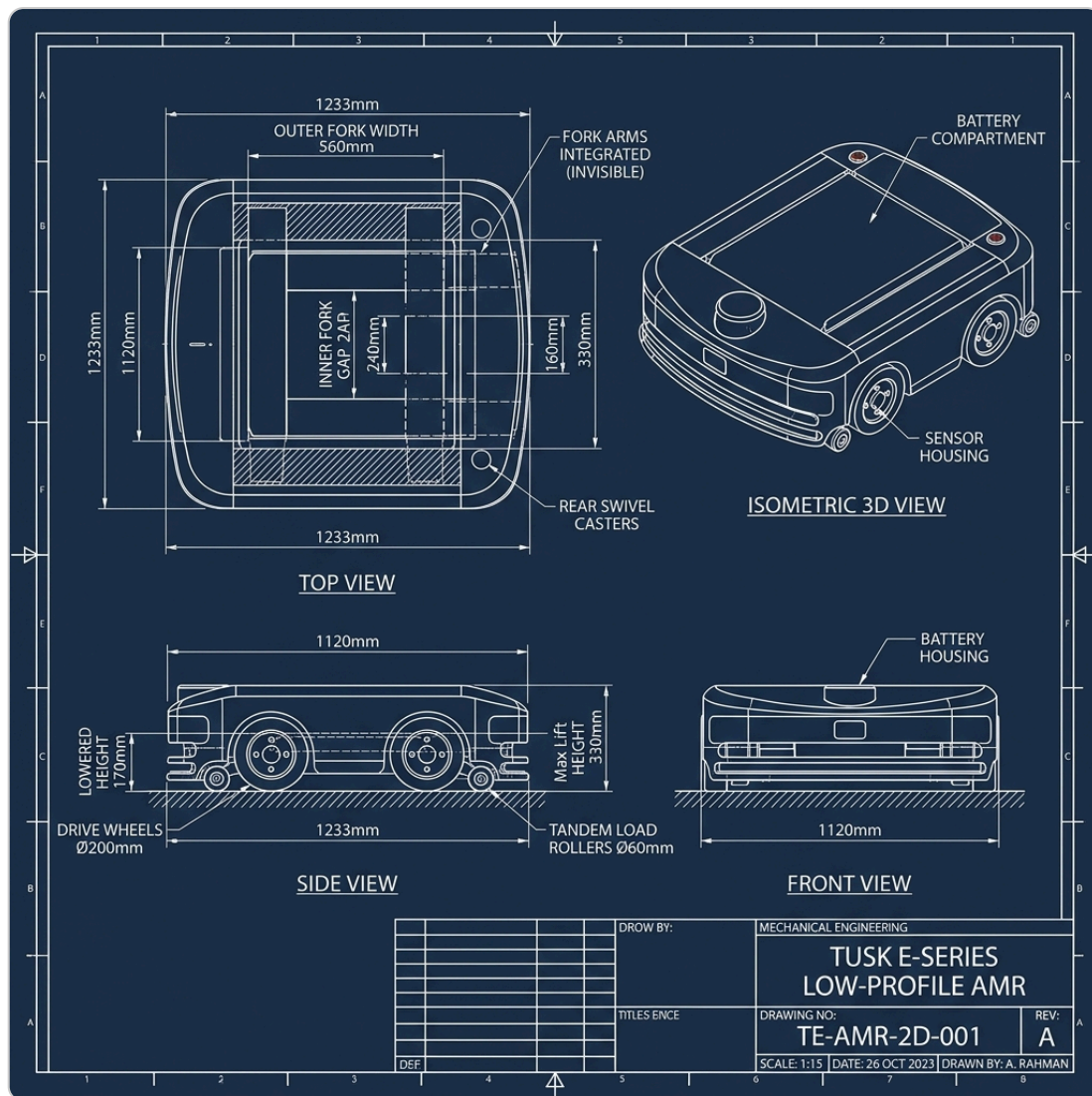
Engineering Parameter	Dimension (mm)	Tolerance	Notes & CAD Reference
Overall Robot Length	1233 mm	$\pm 1.0 \text{ mm}$	Front bumper to rear caster housing
Overall Robot Width	1120 mm	$\pm 1.0 \text{ mm}$	Outer steel weldment frame envelope
Chassis Lowered Height	170 mm	$\pm 0.5 \text{ mm}$	Top of platform to floor (lowered)
Max Lift Height (Top of Deck)	330 mm	$\pm 1.0 \text{ mm}$	Total vertical lift travel: 160 mm

Engineering Parameter	Dimension (mm)	Tolerance	Notes & CAD Reference
Fork Assembly Outer Width	560 mm	$\pm 0.5 \text{ mm}$	Fits standard EUR-2 / ISO pallets
Fork Assembly Inner Gap	240 mm	$\pm 0.5 \text{ mm}$	Space between left and right tyne
Individual Tyne Width	160 mm	$\pm 0.5 \text{ mm}$	$\frac{560 - 240}{2} = 160 \text{ mm}$ tyne width
Fork Nominal Length (E10)	1100 mm	$\pm 1.0 \text{ mm}$	Standard fixed length fork tyne
Fork Extension Reach (E10T)	1400 mm max	$\pm 2.0 \text{ mm}$	Multi-stage telescopic reach
Drive Wheel Diameter	$\varnothing 200 \text{ mm}$	$\pm 0.2 \text{ mm}$	Polyurethane tread, 65 mm width
Tandem Load Roller Diameter	$\varnothing 60 \text{ mm}$	$\pm 0.2 \text{ mm}$	Polyurethane 95A, 180 mm length
Roller Bogie Outer Length	198 mm	$\pm 0.5 \text{ mm}$	Fits 180 mm roller + $2 \times 8 \text{ mm}$ plates
Roller Shaft Specs	$\varnothing 20 \text{ h6} \times 188 \text{ mm}$	$\pm 0.1 \text{ mm}$	EN8 Steel with circlip grooves

Engineering Parameter	Dimension (mm)	Tolerance	Notes & CAD Reference
Minimum Ground Clearance	25 mm	$\pm 0.5 \text{ mm}$	Chassis base plate to floor clearance

2. 2D Master CAD Dimensioned Blueprint Sheet

Use the engineering drawing sheet below for placing sketch origins, centerlines, and datum planes during 3D CAD modeling:



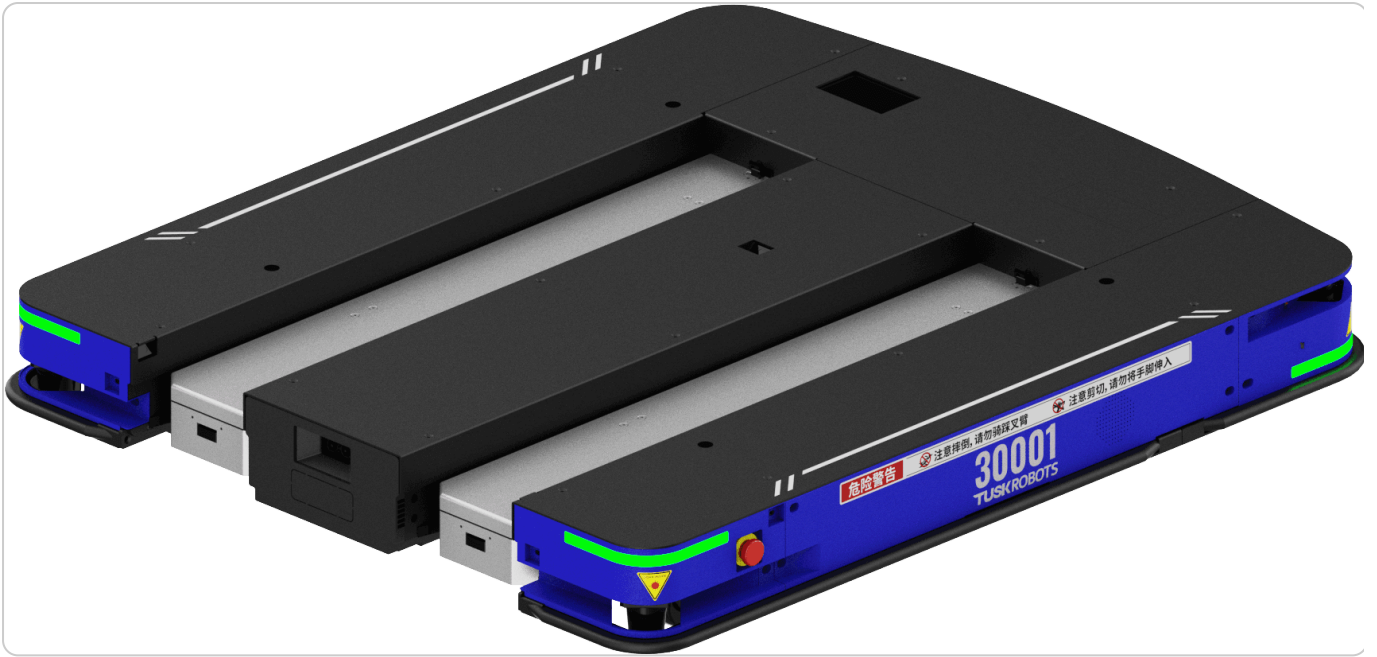
Tusk E-Series Low-Profile Pallet AMR Master 2D CAD Dimensioned Blueprint Sheet (Illustrative Schematic / Non-Realistic)

3. Product Family Real Models Showcase

A. Tusk Model E10 (Standard Low-Profile Pallet AMR)

Integrated low-profile pallet AMR featuring fixed-length "invisible" fork tynes and passive mechanical pull-rod wheel deployment.

- **Payload Rating:** 1000\text{ kg}
- **Real Product Photo:**

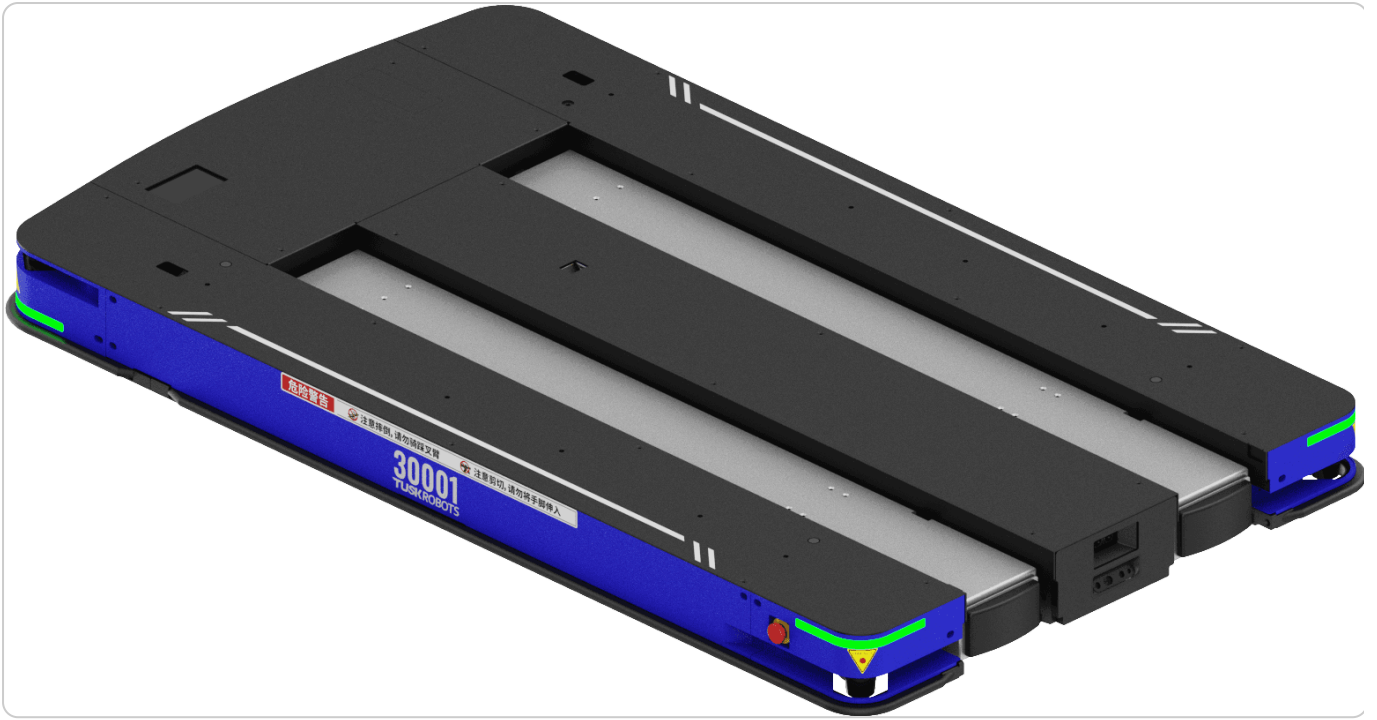


Real Tusk E10 / E10D Standard Low-Profile Pallet AMR Product Photo

B. Tusk Model E10T (Telescopic Extending Fork AMR)

Equipped with dual-stage telescoping tynes that extend up to 1400 mm out into pallet racks while the AMR remains stationary.

- **Payload Rating:** 1000 kg
- **Extension Reach:** 1400 mm
- **Real Product Photo:**



Real Tusk E10T Telescopic Extending Fork Pallet AMR Product Photo

4. Detailed Component Bill of Materials (BOM) & Materials

When modeling individual parts and assemblies, use the following material assignments and component callouts:

Item No.	Component Name	Material / Spec	Quantity	Critical Dimensions / Notes
1	Base Frame Weldment	Q235 Steel Tube ($80\times 60\times 4\text{ mm}$)	1 Set	Inner frame cutout: $960\text{ mm} \times 713\text{ mm}$

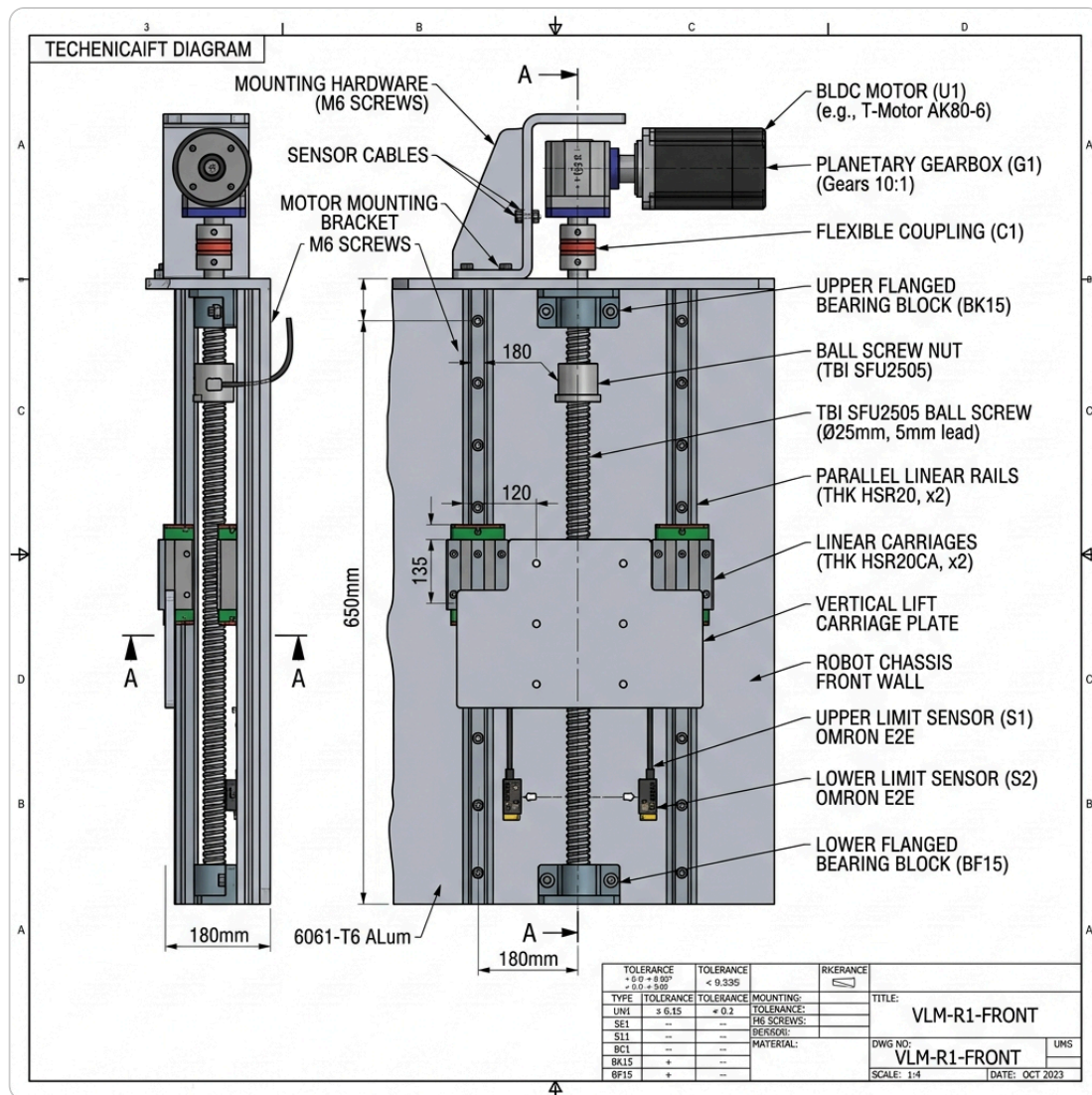
Item No.	Component Name	Material / Spec	Quantity	Critical Dimensions / Notes
2	Top Platform (Fork Deck)	Q235 Steel Plate (8\text{ mm})	1 Pc	Overall: 1100\text{ mm} ; 8\text{ mm} \varnothing 11\text{ mm} holes
3	Scissor Arm (Part 1)	Rectangular Tube (60\text{ mm} \times 40\text{ mm} \times 6\text{ mm})	2 Pcs	Total length: 760\text{ mm} ; Pivot C-to-C: 720\text{ mm}
4	Scissor Arm (Part 2)	Rectangular Tube (60\text{ mm} \times 40\text{ mm} \times 6\text{ mm})	2 Pcs	Mirrored cross arm with central \varnothing 20\text{ mm} bore
5	Cross Pin (Pivot)	EN8 Steel (\varnothing 20\text{ mm})	4 Pcs	Body length: 120\text{ mm} ; Total length: 140\text{ mm}
6	Pivot Bushing	Bronze CuSn12	8 Pcs	Outer: \varnothing 25\text{ mm} , Inner: \varnothing 20\text{ mm} , Length: 25\text{ mm}
7	Ball Screw Shaft	SCM415 Hardened Steel	1 Pc	Spec: SFU2505 (Dia 25\text{ mm} , Lead 5\text{ mm}), 600\text{ mm} L
8	Ball Nut & Housing	Cast Steel Block	1 Pc	Nut body: \varnothing 40\text{ mm} , Flange: 55\text{ mm} sq, 4\text{ mm} \times \varnothing 9 holes

Item No.	Component Name	Material / Spec	Quantity	Critical Dimensions / Notes
9	BK15 Bearing Block	Cast Iron Housing	2 Pcs	Base: $89 \times 34 \text{ mm}$, Height: 39 mm , Bore: $\varnothing 15 \text{ mm}$
10	Drive Motor	750W 24V BLDC Motor	2 Pcs	Rated speed: 3000 RPM , Gearbox ratio: 1:20
11	Lift Motor	1000W 24V BLDC Motor	1 Pc	Rated speed: 3000 RPM , Gearbox ratio: 1:15
12	Roller Bogie Housing	Q235 Steel Plate (8 mm)	2 Sets	Outer length: 198 mm , Height: 102 mm , Width: 75 mm
13	Polyurethane Roller	Polyurethane 95A Shore	4 Pcs	Outer dia: $\varnothing 60 \text{ mm}$, Inner dia: $\varnothing 20 \text{ mm}$, Length: 180 mm

5. Subsystem Mechanical Schematics

A. Vertical Lift Carriage Drive

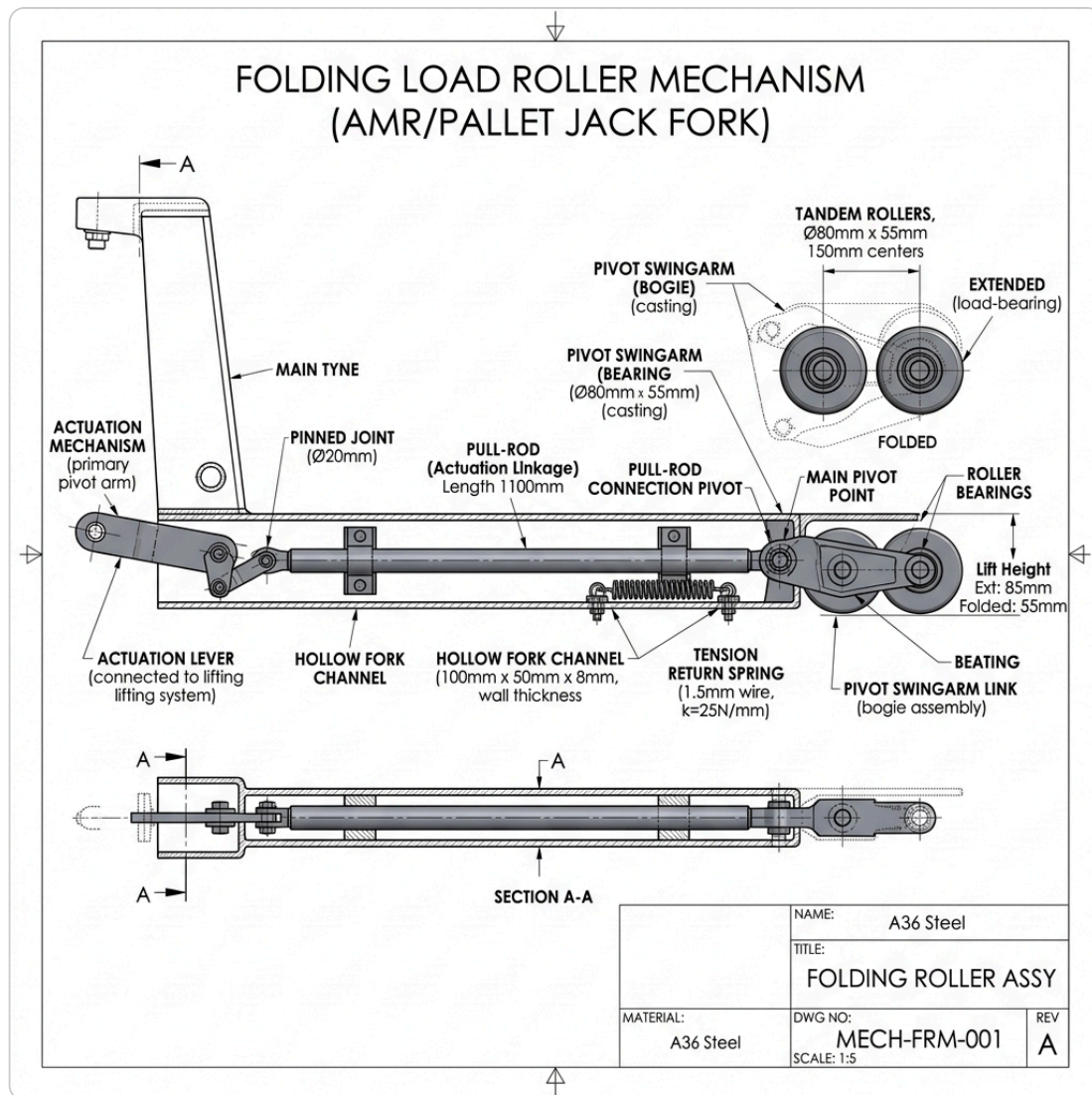
The vertical carriage moves on twin HGR25 linear rails and is driven by the central SFU2505 ball screw:



Vertical Lift Carriage Assembly Schematic Diagram (Illustrative Schematic / Non-Realistic)

B. Folding Load Roller Kinematics

For standard E10 models, the front tandem load rollers retract inside the fork cavity when lowered, and deploy down to touch the floor when raised:



Folding Load Roller Mechanism Schematic Diagram (Illustrative Schematic / Non-Realistic)

6. General CAD Modeling Guidelines & Tolerances

1. Standard Machining Tolerances (ISO 2768-m):

- 0 - 6\text{ mm} : $\pm 0.1\text{ mm}$
- 6 - 30\text{ mm} : $\pm 0.2\text{ mm}$
- 30 - 120\text{ mm} : $\pm 0.3\text{ mm}$
- 120 - 400\text{ mm} : $\pm 0.5\text{ mm}$
- 400 - 1000\text{ mm} : $\pm 0.8\text{ mm}$

- $1000 - 2000\text{ mm}$: $\pm 1.0\text{ mm}$

2. Weldment Notes:

- All structural tube joints require 6 mm continuous fillet welds (MIG process, ER70S-6 filler wire).
- All sharp edges and corners must be deburred with $R1 - R2\text{ mm}$ radii.

3. Surface Finish:

- Industrial powder coating (Orange RAL 2004 / Dark Grey RAL 7016), dry film thickness $80 - 100\text{ }\mu\text{m}$.